MIL-P-20334A (MU)
15 February 1967
SUPERSEDING
MIL-P-20334
21 November 1951
and
PA-PD-2576
24 July 1964

MILITARY SPECIFICATION

PYROTECHNICS, WHITE AND COLOR COMPOSITIONS

1. SCOPE

1.1. This specification covers the ingredients and chemical composition of white and color pyrotechnics compositions.

1.2. Classification. The pyrotechnic white color compositions shall be of the following types as specified (see 6.1):

Type 1 — White and Yellow Composition Table I.

Type 2 — Red Composition, Table II.

Type 3 — Green Composition, Table III.

Type 4 — Yellow and Amber Composition, Table IV.

2. APPLICABLE DOCUMENTS

2.1. The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 — Sieve Standard Testing.

JJJ-O-86 — Oil, Castor, Technical Grade.

JJJ-O-331 — Oil Linseed Boiled.

MILITARY

MIL-P-156 — Potassium Nitrate.

MIL-B-162 — Barium Nitrate.

MIL-S-20322 — Strontium Nitrate Non-Hygroscopic.

MIL-P-223 — Powder, Black.

MIL-P-217 — Potassium, Perchlorate.

JAN-S-210 — Sodium Oxalite (Technical Grade)

JAN-S-487 — Sulfur (For Use in Ammunition)

JAN-H-257 — Hexachlorbenzene.

MIL-A-512 — Aluminum, Flaked, Grained and Atomized (For Use in Ammunition).

JAN-A-356 — Asphaltum (Gilsonate).

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MIL-T-407 — Thermite Plain Incendiary.

JAN-C-768 — Copper, Powder.

MIL-P-20307 — Polyvinyl Chloride.

MIL-C-15198 — Calcium Carbonate Precipitate Pigment.

MIL-A-2550 — Ammunition and Special Weapons; General Specification for.

JAN-D-232 — Dextrin (For Use in Ammunition).

JAN-S-322 — Sodium Nitrate (For Ordnance Use).

JAN-M-382 — Magnesium, Powder (For Use in Ammunition).

MIL-S-20465 — Starch, Potato.

MIL-S-322 — Sodium Nitrate (For Ordnance Use).

STANDARDS

MILITARY

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-650 — Explosive; Sampling, Inspection and Testing.

MIL-STD-1168 — Lot Numbering Of Ammunition.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The raw materials used in the manufacture of pyrotechnic compositions shall conform to the requirements of the applicable specifications stated in table I to IV, except as modified by notes in tables I to IV.

3.1.1 *Coated magnesium.* Magnesium, type 1, grade A, granulation 11, in the various compositions shall be coated entirely with 8 plus or minus 0.1 percent by weight of boiled linseed oil, type A, or with a 50/50 mix of linseed oil and castor oil. Magnesium, type 1, grade A, granulation 4, shall be coated entirely with approximately 6 plus or minus 0.1 percent by weight of the linseed oil or the mix. When the magnesium is uniformly coated, it shall be dried in shallow trays for 48 hours min. at 40 percent RH (max) and 80 degrees F. (max). During drying on the trays the material shall be agitated so as to prevent a crust from forming on the composition. The dried material shall be broken up in any suitable manner so that it will pass through a U.S. Standard No. 20 sieve.

3.1.2 *Moisture content of the material.* The material shall be dried in accordance with the detail specification or as specified in the contract before mixing into a composition.

3.2 Composition. Tables I, II, III, IV, indicate the approximate percentages of the various ingredients to be used for the particular compositions specified on drawings and detail specifications. These percentages shall be adjusted to meet the individual candlepower, burning time and color value requirements of the item as shown on the applicable drawings and detail specifications. All of the ingredients shown in the appropriate table for a particular composition shall be used in its manufacture and only the ingredients specified shall be used.

3.2.1 *Moisture content of the compositions.* The maximum moisture content of the composition shall be in accordance with the detailed specification of the item being manufactured.

3.3 First fire compositions. Unless otherwise specified on drawings or in detail spec-

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ification, the first fire composition for a particular item shall consist of a dry blend of 75 percent of the pyrotechnic composition (see tables I to IV) and 25 percent of black powder class V.

3.4. **Workmanship.** Compositions shall be thoroughly blended to form a homogeneous mixture.

4. QUALITY ASSURANCE PROVISIONS

4.1. **General quality assurance provisions.** Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of this specification including statements of findings resulting from analyses of the composition under contract.

b. Number of batches of product inspected.

c. Results obtained for all inspection performed.

d. Specification numl r and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of pounds in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agents authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using this detail specification, and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor.

4.2 **Inspection provisions.**

4.2.1 *Lot formation.* The term lot as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous quantity of pyrotechnic composition from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of pyrotechnic composition from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more batches of white or color composition produced by

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one manufacturer, in accordance with the same specification, or same specification revision under one continuous set of operating conditions. Each lot shall consist of that quantity of white and color composition which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL–STD–1168. Unless otherwise approved by the contracting officer, the inspection lot size of composition deliverable under the contract shall be not more than the largest quantity contractually scheduled for delivery during any month of the contract period.

4.2.2 *Testing.*

4.2.2.1 *Sampling.* Unless otherwise specified, sampling shall be in accordance with detail specifications of the items being manufactured. At the discretion of the inspector, two 4-ounce samples of the composition shall be placed in separate sealed containers, and marked for identification. One sample shall then be retained by the inspector and the other shall be forwarded to the Commanding Officer, Picatinny Arsenal, Dover, N. J., or a laboratory designated by the Government together with pertinent information for analysis to determine compliance with the requirements of paragraph 3.2.

4.3 **Test methods and procedures.**

4.3.1. *Moisture content.* The moisture content of the composition shall be determined in accordance with the detailed specifications for the item being manufactured.

5. PACKING (Not applicable.)

6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL–STD–1167.

6.2 Unless otherwise specified it is advisable that all operations in connection with the handling, blending, mixing, consolidation, and storage of compositions be carried out in a room in which the relative humidity is no greater than 40 percent as determined by periodic readings with a sling psychrometer. Compositions shall be maintained under the specified controlled conditions for use within 8 hours after completion of blending. Compositions may be stored in hermetically sealed containers for use within 32 hours after the original blending, but must be destroyed after this period. Compositions that are stored should be adequately marked for identification.

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TABLE I. *White and Yellow Tinted Compositions Type 1*

(Approximate percentages (by weight) of ingredients required.)

1	Ingredient	Coated Magnesium JAN-M-382		Aluminum MIL-A-512		Barium Nitrate MIL-B-162	Sodium Nitrate MIL-S-322	Strontium Nitrate MIL-S-20322	Sulfur JAN-S-487	Sodium Oxalate JAN-S-210	Asphaltum JAN-A-356	Calcium Carbonate MIL-C-13198	Castor Oil JJJ-C-86	Linseed Oil JJJ-O-331	Dextrin JAN-A-232	Thermite MIL-T-407
2	Type, Grade, Class of Granulation	I-A-11	I-A-4	I-B-3	II-E-6	Class 6 Gran A		A	B	a				A	II	
3	Composition 101		20		19	34				25			1	1		
4	101A		36		8	36				18			1	1		
5	101B		40	9		37				12			1	1		
6	101C		43		11	38				7			0.5	0.5		
7	101D		38		7.5	38				14			1	1.5		
8	101E		46		8	36				8			1	1		
9	101F		39		7.5	37				15.5			0.5	0.5		
10	102				26	66			6				2			
11	103			2	20	62		11	3					2		
12	104	18			15	57		5			2		1.5	1.5		
13	105		17		15	55		5			5			3		
14	106			14	10	59			4	11			1	1		
15	107			16.2	10.1	52.6	10.1		4	5			1	1		
16	108				21	68			4	5			2			
17	109		33				35			16	5	10	0.5	0.5		
18	110		15		18	51		6			9				1	
19	111		38 (a)				43 (b)				2	14 (c)	3			
20	112				6.5	38			8.5				0.5			46.5
21	113				17	34.5			1.5				0.5			46.5
22	114				18.5	25.5			2				0.5			53.5

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TABLE II. *Red Compositions Type 2*

(Approximate percentages (by weight) of ingredients required.)

1	Ingredient	JAN-M-382 Coated Magnesium		Potassium Nitrate MIL-P-156	Potassium Perchlorate MIL-P-217	Strontium Nitrate MIL-S-20322	Hexachlor-Benzene JAN-H-257	Polyvinyl-Chloride MIL-P-20307	Asphaltum JAN-A-358	Linseed Oil JJJ-O-331	JAN-D-232 Dextrin	Paraffin Oil	Potato Starch MIL-S-20485
2	Type, Class, Grade, or Granulation	I-A-11	I-A-4	Class II	A or B Class 4	A				A	II	See Note (d)	
3	Composition 201	34 (e)			25	30	5		5	0.5			
4	201A	10			20	55	5		10				
5	201B	13			20	54	8		5				
6	201C	18 (e)			24	44.5	6		7	0.5			
7	201D	45.5			29	18	4		3	0.5			
8	201E		33		21	31	5		10				
9	201F	21			15	45	12		7				
10	201G	32			20	32	8		8				
11	202	18			24	45	6			2	5		
12	202A	34.4			25.3	30.3	5			2	3		
13	203	25			26	35	6		7	1			
14	203A		21		15	52	5		6	1			
15	204	46			29	18	4				1	2	
16	205	36				54 (f)	5		5				
17	206	21.2		19.2		48.6	6		7				
18	206A	37.6		13.8		37.6	6		5				
19	206B	49		15		28	5		3				
20	207		30		29	33	4		2			2	
21	208	30 (a)				54		8	8				
22	209	18			24	45		6	7				
23	210		30 (a)			62		4	2			2	
24	211		31		19	28	4		8				10
25	211A		40		22	18	6		7				7
26	211B		25.5		19.5	28	4.5		9				13.5

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TABLE III. *Green Compositions Type 3*

(Approximate percentages (by weight) of ingredients required.)

1	Ingredient	JAN-M-382 Coated Magnesium		Barium Nitrate MIL-B-162	Potassium Perchlorate MIL-P-217	JAN-C-768 Copper Powdered	Hexachlor-Benzene JAN-H-257	Polyvinyl-Chloride MIL-P-20307	Asphaltum JAN-A-356	Linseed Oil JJJ-O-331	Castor Oil JJJ-C-86	JAN-D-232 Dextrin	Paraffin Oil
2	Type, Class, Grade or Granulation	I-A-11	I-A-4	Class 6 Gran A	A or B Class 4					A		II	See note (d)
3	Composition 301	16		59		2	21			2			
4	301A	12		62		2	21			3			
5	301B	15		66		2	15			2			
6	302	16 (e)		51	12	1	12			3		5	
7	303	48		16	25	2	6		3				
8	303A	27		48	9	2	12		2				
9	304	48		16	25	2	6					1	2
10	305	27		48	9	2	12			1		1	
11	306	20		21	36		16		5	1	1		
12	307		16	60		2	21		1				
13	307A		23	53		2	20		2				
14	307B		27	46		2	23		2				
15	307C		16	59		2	21		2				
16	308	21		49	10			16	2	2			
17	309		15	27	36		16		5	1			
18	310		26 (a)	45	16	2	7		2				2
19	311		26 (a)	16	45	1		7	3				2

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TABLE IV. *Yellow and Amber Compositions Type 400*

(Approximate percentages (by weight) of ingredients required.)

1	Ingredient	JAN-M-382 Coated Magnesium		Barium Nitrate MIL-B-162	Potassium Perchlorate MIL-P-217	Strontium Nitrate MIL-S-20322	Sodium Oxalate JAN-S-210	Sodium Resinate	Hexachlor-Benzene JAN-H-257	Polyvinyl-Chloride MIL-P-20307	Asphaltum JAN-A-356	Linseed Oil JJJ-O-331	Dextrin JAN-D-232	Paraffin Oil
2	Type, Class, Grade or Granulation	I-A-11	I-A-4	Class 6 Gran A	A or B Class 4	A	a	See Note (g)				A	II	See Note (d)
3	Composition 401	23 (e)			50		13	10	2			2		
4	402	20			45		22		6		6	1		
5	402A	30			39		13		7		9	2		
6	402B	10			50		16		9		12	3		
7	402C	15 (e)			50		16		8		9	2		
8	402D	13			44		26		10		5	2		
9	403	19 (e)			49		15	12	5					
10	403A	23			50		13	10	4					
11	404	44		14	28	6	2		3				1	2
12	405	44		14	28	6	2		3		3			
13	406	26		4	42		15		6		7			
14	407	26		4	42		15		6			2	5	
15	408	15 (e)			50		16		8			4	7	
16	409	17		24	15	8	16		11			3	6	
17	410	30		39			13		7		9	2		
18	411	20			50		16	12		2				
19	412	19			47.6		15.3			7.6	8.6	1.9		
20	413		27	45			23					1		
21	414		26 (a)	29	23		13		5		2			2
22	415		26 (a)		52		13			5	2			2
23	416		34.5		24.5		19				11			

Notes. Comp. 413 – Add 4 percent aluminum type EC in accordance with MIL-A-512.
Comp. 416 – Add potato starch 11 percent in accordance with MIL-S-20463.

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Notes. Tables I, II, III, IV:

(a) Apparent density (minimum) of the magnesium for compositions 111, 208, 210, 414 and 415 shall be 0.56. For compositions 310 and 311 – 0.90 minium.

(b) Granulation of sodium nitrate for composition 111 is preferred as 60 percent maximum through a No. 200 U. S. Standard sieve.

(c) Apparent density of calcium carbonate for composition 111 in the range 17 to 30 gm. per 100 ml is preferred (determination by the method No. 201.1 of Standard MIL–STD–650).

(d) Paraffin oil shall have the following properties:

Gravity (API)	28–32
Flash Point (min).	320°F.
Fire Point (min)	365°F.
Pour Point (max)	25°F.
Color (NPA) (max)	1½
Viscosity (S. U. at 100°F.)	70–75

(e) Either granulation 4 or 11 of Type I, Grade A, magnesium may be used. Both have been found satisfactory.

(f) Use Strontium Nitrate, Grade A.

(g) Technical grade of commercial sodium resinate.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. *This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.* Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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